

MUSA
MAHARASHTRA UNIVERSITY
STUDENTS ASSOCIATION

PRESENTS

CODEX
— 2 0 2 6 —

Unfiltered Devs

Selected Domain Name: Artificial Intelligence & Machine Learning

Selected Problem Statement: The RTI Black Hole

Selected PS No.: CX0107

Team Name: Unfiltered Devs

Team Details and Selected Theme

Team Name =

- Member 1 Name –** Ajay Yadav – Team Lead & Frontend Developer
Member 2 Name – Rangesh Gupta – AI/ML & Backend Developer
Member 3 Name – Shivam Prajapati – UI/UX & Frontend Developer
Member 4 Name – Poornita Sahu – Research, Presentation

Selected Theme =

Artificial Intelligence & Machine Learning

- Why we selected:–** We selected this approach to handle multilingual RTI applications efficiently. It combines OCR, language detection, and AI-based department classification. Jurisdiction rules and confidence checks help prevent incorrect routing. Human escalation ensures ambiguous cases are handled safely and transparently.

Problem Statement & Target Users

WHAT PROBLEM ARE WE SOLVING?

- RTI applications are often **multilingual, scanned**, and difficult to **classify and route** accurately.
- Public authorities and RTI officials face **delays** and **errors** in identifying the correct department and jurisdiction.
- Accurate routing is important to ensure **faster and more transparent** processing of RTI applications.
- If unresolved, applications may be **misrouted, delayed**, or require repeated **manual intervention**.

PRIMARY USERS

-  **RTI applicants / citizens**
(People seeking information)
-  **RTI officers and public authorities**
(Handling and processing applications)
-  **Government departments**
(Receiving and responding to RTI applications)



Existing Solutions & their Limitations

Existing Solutions

01 Solution 1: Manual Routing

RTI applications are manually reviewed by officials and routed to the relevant department based on their experience.



02 Solution 2: Keyword-Based Systems

Some departments use simple keyword matching or rule-based systems to categorize and route RTI applications.



03 Solution 3: Basic OCR Tools

Standalone OCR tools are used to convert scanned documents into text, but they often struggle with multiple languages and poor-quality images.



Limitations / Gap

01 Manual Routing Limitations

- Slow and time-consuming
- Prone to human error
- Inconsistent decision-making



02 Keyword-Based System Limitations

- Misses context and intent
- Fails with synonyms and variations
- Limited to single language or keywords



03 Basic OCR Limitations

- Low accuracy with poor image quality
- Struggles with multilingual documents
- Loses layout and structure



The Gap

No existing solution handles multilingual, context-aware, and intelligent routing with jurisdiction checks and human escalation.

Proposed key and Key Innovation

Our Proposed Solution

ROI Backhole

An AI-powered, multilingual intelligent RTI document routing system that classifies, understands and routes applications to the right department — automatically.

How It Works

- 01 Document Ingestion**
Accepts RTI applications in PDF, image or other formats and detects the language (10+ Indian languages).
- 02 AI-Powered Understanding**
Extracts key information, identifies intent, classifies the request and detects jurisdiction using NLP and ML models.
- 03 Smart Routing**
Matches the application with the correct department and jurisdiction, and routes it automatically.
- 04 Tracking & Feedback**
Provides real-time status, ensures transparency and learns from feedback for continuous improvement.



Key Innovation

What makes it different?



Multilingual Intelligence

- Understands 10+ Indian languages (text + scanned documents).
- No language barrier for citizens.



Jurisdiction-Aware Routing

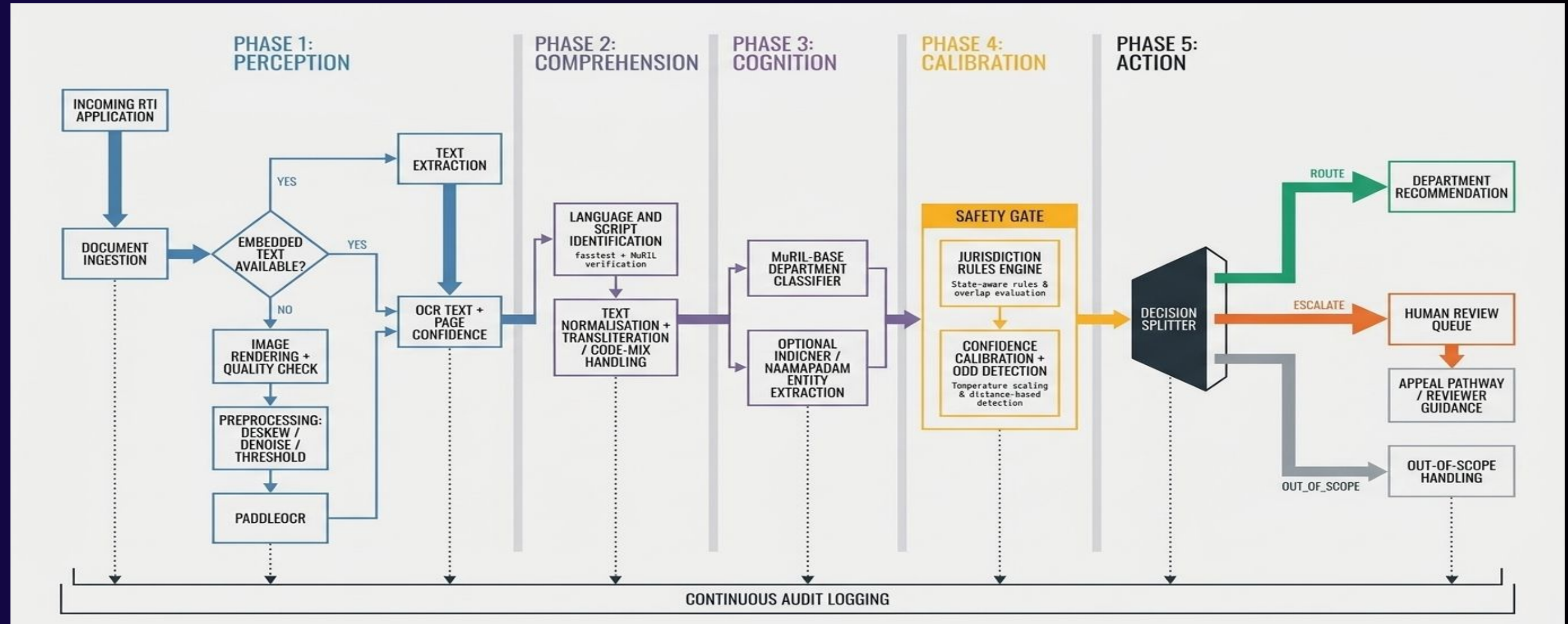
- Uses location, department rules & content understanding to find the right authority.
- Reduces manual effort and misrouting.



Continuous Learning

- Learns from past decisions and officer feedback.
- Improves accuracy over time.

Technical Approach / System Workflow



Feasibility & Expected Impact

Technical Feasibility

Designed to run locally on standard workstations with modest hardware requirements. It uses lightweight, encoder-style components to avoid cloud dependency and support district-level deployment.

Safety Gates

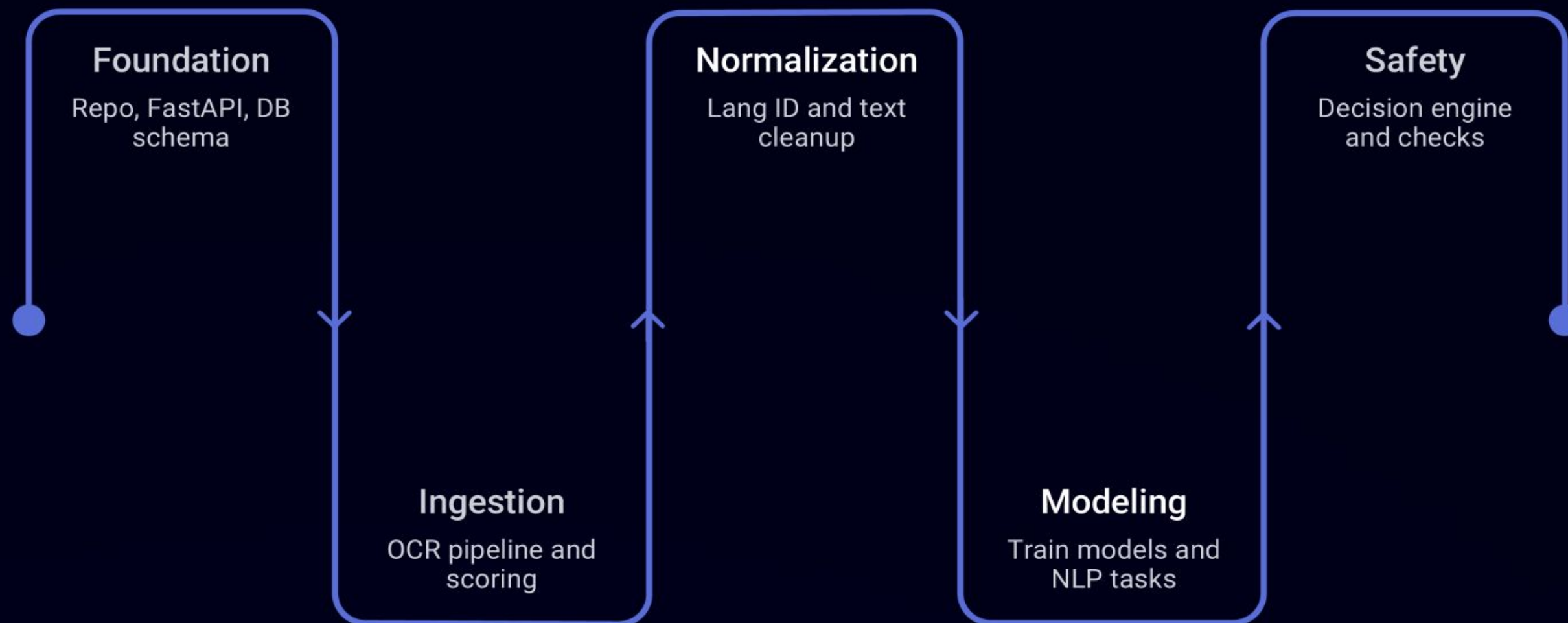
Temperature scaling, OOD rejection, and OCR quality thresholds enforce the Zero False-Certainty Principle. Ambiguous cases are escalated instead of being routed blindly.

Operational Benefits

Straight-through routing reduces first-read loops and frees officers for complex cases. It also shortens citizen wait times while preserving human oversight.

Development Plan

Development Roadmap — 8 Phases to Production



The roadmap moves from core infrastructure and OCR ingestion through normalization, modeling, and safety controls. It then adds review tooling, dashboards, and final evaluation to validate the system for production.

Solution Summary / Expected Outcome

Solution Summary – Key Takeaways and Impact

Automated, Multilingual Accuracy

Leverages OCR, language ID, and state-aware routing to achieve high accuracy across 8 Indian languages, dramatically reducing misclassification and processing delays.

Boosted Operational Efficiency

Streamlines RTI filing routing, minimizes manual first-read loops, and frees up human resources for more complex cases, leading to shorter citizen wait times.

Secure & Fully Auditable

Ensures safe processing with a "Zero False-Certainty" principle, escalating ambiguous cases for human review, and maintaining a complete, traceable audit trail for every action.

Offline-First Deployment

Designed for local deployment on standard workstations in district offices, ensuring continuity and accessibility even without constant cloud connectivity.

References & Disclosure of Existing Work

01

RTI Online

Official RTI filing & authority reference

<https://rtionline.gov.in/>

02

RTI-Bench

RTI-domain dataset & research reference

<https://github.com/joyboseeroy/rti-bench>

03

RTI Wiki

Cases, terminology & sample applications

<https://righttoinformation.wiki/cases>

04

Government Open Data

RTI datasets & government metadata

<https://www.data.gov.in/keywords/RTI>

05

MuRIL

Multilingual NLP / routing model

<https://huggingface.co/google/muril-base-cased>

06

IndicBERT

Multilingual benchmark model

<https://huggingface.co/ai4bharat/indic-bert>

07

IndicTrans2

Indian-language translation resource

<https://github.com/AI4Bharat/IndicTrans2>

08

fastText

Language identification resource

<https://fasttext.cc/docs/en/language-identification.html>